

JEE MAIN REVISION MASTER: GASEOUS STATE OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 63 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Gaseous State — Standard Calculations & Problem Archetypes	36 PYQs	Quantitative Analysis
Very High	Gaseous State — Core Principles & Governing Laws	17 PYQs	Core Principle
High	Gaseous State — Applied Cases, Reactions & Traps	13 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Gaseous State — Standard Calculations & Problem Archetypes	$V = \frac{2 \times 0.083 \times 300}{2 \times 1} = 24.9 \text{ litre}$, $\frac{RT}{P}$, $PV = \frac{W}{M}RT$
Gaseous State — Core Principles & Governing Laws	0.0125 m^3 , a^b , 10^{-1} atm
Gaseous State — Applied Cases, Reactions & Traps	$\frac{PM}{RT}$, CH_4 , $\frac{2}{R}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Gaseous State — Standard Calculations & Problem Archetypes (Very High Yield — 36 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Gaseous State.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2025 Morning Shift] Q#1 (MCQ): "At the sea level, the dry air mass percentage composition is given as nitrogen gas: 70.0, oxygen gas: 27.0 and argon gas: 3.0. If total pressure is 1 atm, the partial pressure of nitrogen gas is..." [Ans: N/A]
- [2023 Morning Shift] Q#3 (MCQ): "For 1 mol of gas, the plot of pV vs. p is shown below. p is the pressure and V is the volume of the gas. What is the value of compressibility factor Z at point A?" [Ans: N/A]
- [2023 Evening Shift] Q#4 (Numerical): "The number of statement/s, which are correct with respect to the compression of carbon dioxide from point (a) in the Andrews isotherm from t_1 to t_2 is..." [Ans: 2]

3.2 Gaseous State — Core Principles & Governing Laws (Very High Yield — 17 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Gaseous State.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Gaseous State.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Gaseous State PYQs.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (Numerical): "Three bulbs are filled with CH_4 , CO_2 and Ne as shown in the picture. The bulbs are connected through pipes of zero volume. When the system reaches equilibrium, the total pressure is 4 atm. The pressure of CH_4 is..." [Ans: 3]
- [2023 Morning Shift] Q#3 (Numerical): "A 300 mL bottle of soft drink has 0.2 M CO_2 dissolved in it. Assuming CO_2 behaves as an ideal gas, the volume of the dissolved CO_2 is..." [Ans: 1362]
- [2022 Morning Shift] Q#2 (Numerical): "The pressure of a moist gas at $27^\circ C$ is 4 atm. The volume of the container is doubled at the same temperature. The new pressure of the dry gas is..." [Ans: 22]

3.3 Gaseous State — Applied Cases, Reactions & Traps (High Yield — 13 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Gaseous State.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Evening Shift] Q#2 (MCQ): "Arrange the following gases in increasing order of van der Waals constant 'a'. A. Ar B. CH_4 C. H_2O D. C_6H_6 Choose the correct order..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (Numerical): "The total pressure of a mixture of non-reacting gases X (0.6 g) and Y (0.45 g) in a vessel is 740 mm of Hg. The partial pressure of the gas X is..." [Ans: 555]
- [2022 Evening Shift] Q#2 (Numerical): "A rigid nitrogen tank stored inside a laboratory has a pressure of 30 atm at 06:00 am when the temperature is $27^\circ C$. At 03:00 pm, the pressure is 32 atm. The temperature at 03:00 pm is..." [Ans: 32]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 7th April Morning Shift (Q#1)

At the sea level, the dry air mass percentage composition is given as nitrogen gas: 70.0, oxygen gas: 27.0 and argon gas: 3.0. If total pressure is 1.15 atm, then calculate the ratio of following respectively: (i) partial pressure of nitrogen gas to partial pressure of oxygen gas (ii) partial pressure of oxygen gas to partial pressure of argon gas (Given: Molar mass of N, O and Ar are 14, 16 and 40 g mol⁻¹ respectively.)

- (A) 5.46, 17.8
- (B) 2.96, 11.2
- (C) 4.26, 19.3
- (D) 2.59, 11.85

Correct Answer: None

Solution Analysis: Partial Pressure Ratio of Nitrogen to Oxygen : The partial pressure of a gas is calculated using its mole fraction. The mole fraction is the ratio of the number of moles of the gas to the total number of moles of all gases. For nitrogen (N₂), the mole fraction $\frac{n_{\text{N}_2}}{n_{\text{N}_2} + n_{\text{O}_2} + n_{\text{Ar}}}$ can be simplified as $\frac{70/28}{70/28 + 27/32 + 3/40}$. For oxygen (O₂), the mole fraction is $\frac{27/32}{70/28 + 27/32 + 3/40}$. Therefore, the ratio of the partial pressure of nitrogen to oxygen is : $\frac{P_{\text{N}_2}}{P_{\text{O}_2}}$

Benchmark Problem 2: JEE Main 2023 (Online) 8th April Morning Shift (Q#1)

Three bulbs are filled with CH₄, CO₂ and Ne as shown in the picture. The bulbs are connected through pipes of zero volume. When the stopcocks are opened and the temperature is kept constant throughout, the pressure of the system is found to be _____ atm. (Nearest integer)

Correct Answer: 3

Solution Analysis: Dalton's law of partial pressure states that whenever two or more gases, which do not react chemically, are enclosed in vessel, the total pressure is equal to sum of partial pressure of each gas. From Dalton's partial pressure law, $P_{\text{f}}V_{\text{f}} = P_{\text{1}}V_{\text{1}} + P_{\text{2}}V_{\text{2}} + P_{\text{3}}V_{\text{3}}$ $P_{\text{f}} \times 9 = 2 \times 2 + 4 \times 3 + 3 \times 4$ $P_{\text{f}} = (28/9) = 3.11 = 3$

Benchmark Problem 3: JEE Main 2023 (Online) 8th April Evening Shift (Q#2)

Arrange the following gases in increasing order of van der Waals constant 'a'. A. Ar B. CH₄ C. H₂O D. C₆H₆
Choose the correct option from the following.

- (A) , C, B and A
- (B) , B, C and D
- (C) , C, D and A
- (D) , D, B and A

Correct Answer: None

Solution Analysis: The van der Waals constant 'a' is a measure of the strength of the intermolecular forces in a gas. Larger molecules and molecules with stronger intermolecular forces will have larger 'a' constants. Here, we are comparing argon (Ar), methane (CH₄), water (H₂O), and benzene (C₆H₆). Argon (Ar) is a noble gas and is monoatomic. Methane (CH₄) is a small molecule with weak intermolecular forces. Water (H₂O) is a small molecule but has strong hydrogen bonding, making its intermolecular forces stronger than those in methane. Benzene (C₆H₆) is a larger molecule than the others and has relatively strong